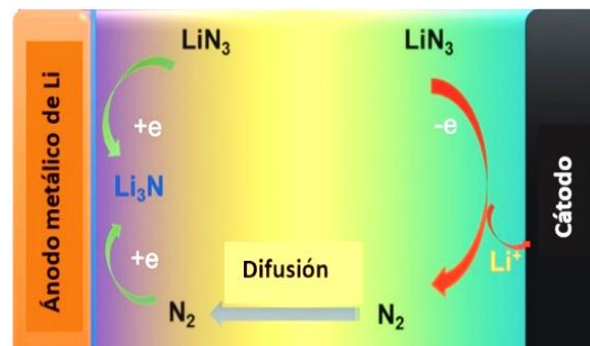


CÓDIGO:

Question 11. Lithium azide (8% of total; only 11th grade)

11.1	11.2	11.3	11.4	11.5	Total	Total
7	7	7	2	9	32	8%

Many researchers in the field of electrochemistry consider that lithium-sulfur batteries may be a new generation of rechargeable lithium batteries due to their high energy density. However, one of the fundamental problems in its design is the stability of the $\text{Li}^{(0)}$ metal anode. Lithium azide LiN_3 ($M = 48.97 \text{ g/mol}$) has been proposed as a possible electrolyte additive in solid-state Li-S batteries. This compound forms a passive layer of Li_3N nitride, which is thin, compact, and has excellent conductivity and covers the lithium anode. The coating of the electrode prevents its deterioration due to the formation of dendrites and the action of polysulfides, increasing the efficiency of the battery and its useful life. The figure schematically illustrates the operation of this type of battery. (Angew. Chem. Int. Ed. 2017, 56(48), 15368-15372; DOI: 10.1002/anie.201709305).



11.1. Represent the Lewis structure and the resonant structures of the azide ion. Make the electronic distribution (of nl^x type) according to the Valencia Bond Theory using hybrid orbitals for the central atom. Consider the formal charges of atoms.

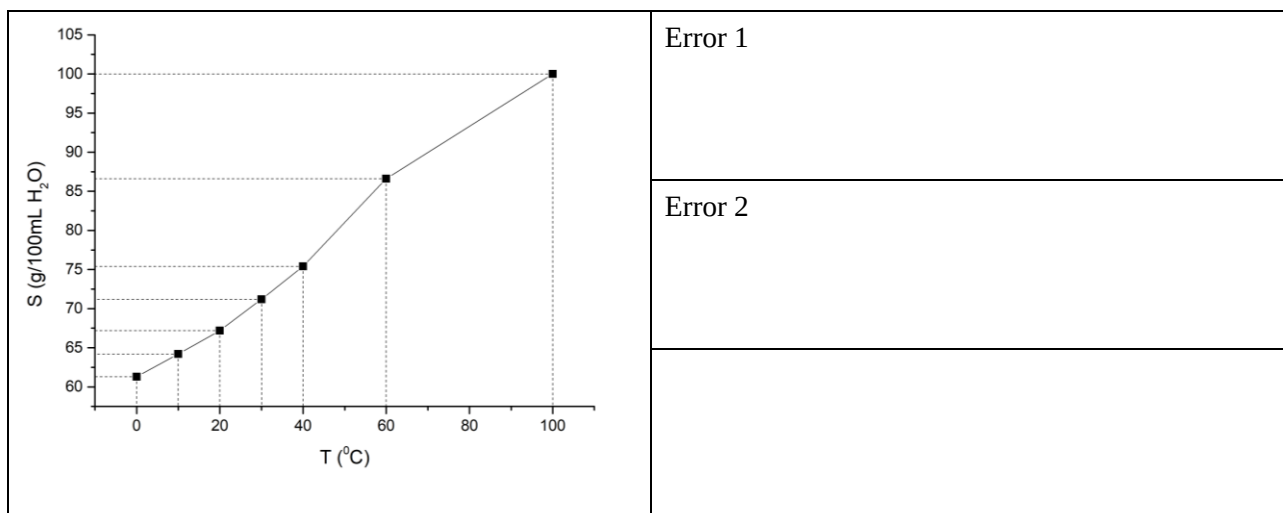
For the design of a battery, a dilute aqueous solution of lithium azide was prepared at 2.0% (m/V) from a more concentrated initial solution of molality 14.0 mol/kg and density 0.707 g/cm^3 at room temperature (25°C).

11.2. Calculate the volume (in μL) of the initial solution required to prepare 125 μL of the 2% solution. Consider that 1% (m/V) = 1 g / 100 mL. $1 \mu\text{L} = 10^{-6} \text{ L}$, $1 \text{ mL} = 1 \text{ cm}^3$

The solubility curve of lithium azide in water is shown below. Use the data provided in the graph to answer the following questions.

11.3. Classify the initial 14 mol/kg solution according to the degree of saturation. Justify numerically.

11.4. Name two errors made during the construction of the graph.



CÓDIGO:

11.5. 100 mL of the initial 14 mol/kg solution was heated until approximately half of the solvent present had evaporated. It was then cooled to 0°C and filtered under reduced pressure. What amount of solid lithium azide could recover after the procedure?

Question 12. First transition series (12% of the total; only 11th grade)

12.1	12.2	12.3	12.4	12.5	12.6	12.7	12.8	12.9	12.10	12.11	12.12	Total	Total
5	2	6	2	2	2	4	3	4	10	6	2	48	12%

12.1. Zinc in an acid medium is an excellent reducing agent, capable of reducing the oxidation state of the ions of the first transition series in aqueous solution. Fill in the colors of the ionic species formed during these processes. (In the table, the symbol X means instability).

Estado	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
VII	---	---	---	---		---	---	---	---	---
VI	---	---	---	orange	green	X	---	---	---	---
V	---	---	yellow	---	X	---	---	---	---	---
IV	---	colorless	blue	---	brown	---	---	---	---	---
III	colorless	purple	green		X		X	X	X	---
II	---	X	purple							
I	---	---	---	---	---	---	---	---	red	---

12.2. Throughout the series, the oxidizing power of the species increases. Because?

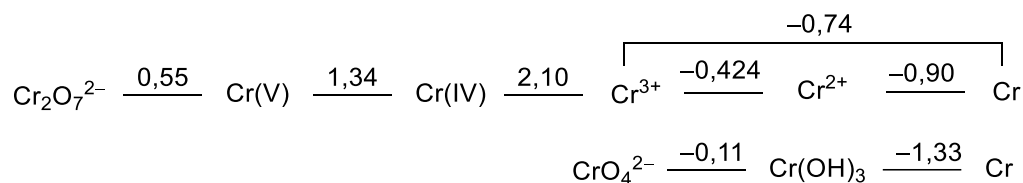
12.3. The ferrate(VI), cobalt(III) and nickel(III) ions are unstable in an aqueous solution and cause the evolution of gas **A**. On the other hand, titanium(II) ions are also unstable in aqueous solution but cause the evolution of gas **B**. Identify both gases and propose a chemical equation that exemplifies the formation of each one.

12.4. Why do ions in oxidation states IV-VII exist as oxygenated cations/anions?

12.5. Titanium(IV) chloride is a liquid at room temperature, while titanium chlorides in lower oxidation states are solids. Explain this difference between the aggregation states.

12.6. Why are the oxides and hydroxides of the above chemical elements generally basic, amphoteric or acidic when the oxidation states are low, intermediate or high, respectively?

Latimer diagrams allow you to obtain relevant information. Use the diagrams of chromium in acidic and basic medium (E° in V, $T = 25^\circ\text{C}$) to answer the following questions.



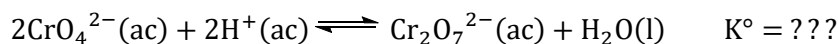
12.7. Which chromium species are unstable in aqueous solution?

$E^\circ(\text{O}_2/\text{H}_2\text{O}) = 1,23 \text{ V (pH} = 0\text{)}; 0,40 \text{ V (pH} = 14\text{)}$ $E^\circ(\text{H}_2\text{O}/\text{H}_2) = 0,00 \text{ V (pH} = 0\text{)}; -0,82 \text{ V (pH} = 14\text{)}$

12.8. Show that the standard potential for the redox couple $\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}$ is 1,33 V.

12.9. Show that the solubility product constant of chromium(III) hydroxide is $1,2 \cdot 10^{-30}$.

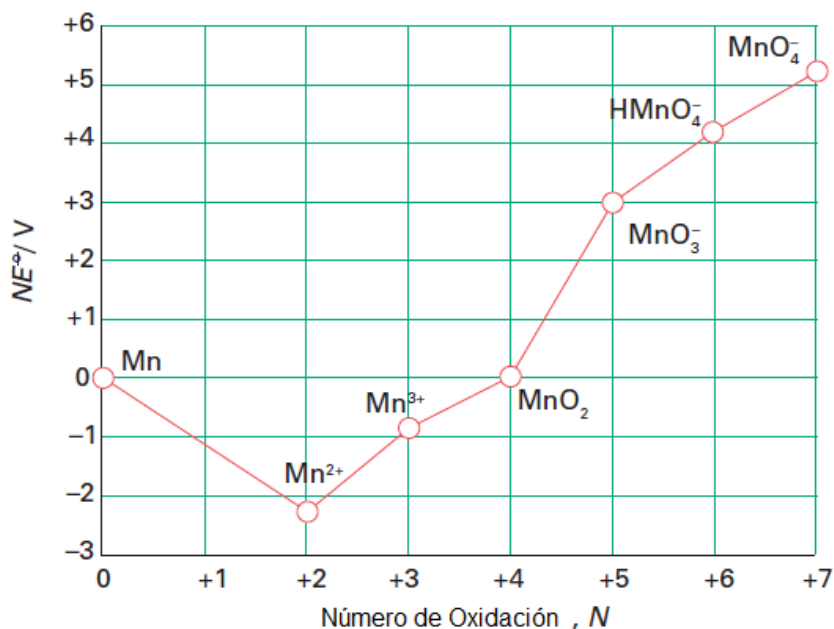
12.10. Knowing that the ionic product of water is $K_w = 1,00 \cdot 10^{-14}$ at 25°C , calculate the formation constant of the dichromate ion from the condensation of two chromate ions in an acid medium:



CÓDIGO:

The Frost diagrams of an element. This type of representation is handy in determining which species are good oxidizing or reducing agents, in addition to providing the possibility of quickly identifying the thermodynamic stability of a given species under certain conditions.

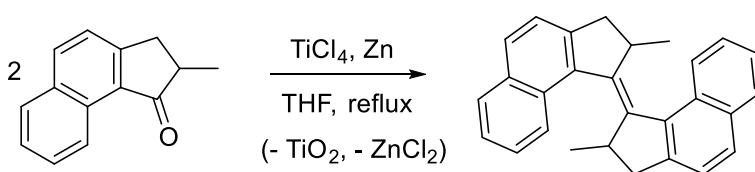
Use the Frost diagram of manganese referenced to the H^+/H_2 redox couple in an acidic solution ($\text{pH} = 0$) to answer the following questions.



12.11. Which species of manganese meet the following characteristics?

Stronger oxidant:	Stronger reductant:
More thermodynamically stable in a dihydrogen atmosphere:	
More thermodynamically stable in a dioxygen atmosphere:	
Unstable with respect to disproportion:	
They combine to form MnO_2 :	
They combine to form Mn^{2+} :	
Insoluble in acidic aqueous solution:	

The above metals are widely used in organic reactions. For example, the McMurry reaction uses a salt of titanium and zinc to form alkenes. Nobel Laureate Ben L. Feringa used this reaction to synthesize first-generation molecular motors (J. Am. Chem. Soc. 2003, 125(49), 15076-15086; DOI: 10.1021/ja036782o). Initially, the solution of the reactants was colorless until the zinc powder was added and the reaction began to occur.



THF: tetrahydrofuran, colorless organic solvent

12.12. About McMurry's reaction, please respond.

i) What is the function of zinc?

ii) Why are Ti(IV) species colorless, while Ti(II) and Ti(III) species are colored?

CÓDIGO:

Question 13. Water in the Swiss Alps (15% of the total; 11th and 12th grades)

13.1	13.2	13.3	13.4	13.5	13.6	13.7	13.8	13.9	13.10	Total	Total
5	10	4	3	6	6	8	10	6	2	60	15%

When Erik and César went to the International Chemistry Olympiad last July, they discovered glaciers on Mount Säntis, more than two kilometers above sea level. To their surprise, the ice remained intact during the mentors' visit to the mountain, even though the temperature was slightly above 0 °C. Intrigued by this discovery, they decided to dust off Peter Atkins' old Physical Chemistry book that was abandoned on the bookshelf and scrutinize its pages in search of answers.

First, they found that atmospheric pressure decreases with increasing height h with respect to sea level according to the barometric equation. In the case of the troposphere, where the temperature varies approximately linearly with a ratio $L_b = 6.5 \text{ K} / 1000 \text{ m}$.

Variation of pressure with height in the troposphere

$$P' = P_b \cdot \left[\frac{T_b - h \cdot L_b}{T_b} \right]^{\frac{g^\circ \cdot M}{R \cdot L_b}}$$

where P is the pressure at height h , above sea level; $P_b = 101.325 \text{ kPa}$ and $T_b = 288.15 \text{ K}$ are the reference pressure and temperature, respectively; $g^\circ = 9.80665 \text{ m/s}^2$ is the acceleration of Earth's gravity; $M = 0.0289644 \text{ kg/mol}$ is the average molecular mass of air; and R is the gas constant.

13.1. Calculate the height (in m) of Mount Säntis if the barometric pressure P' at the summit is 74.7 kPa.

Then, Erik came up with an equation that allows us to calculate the phase transition temperature when the pressure differs from the standard pressure.

Clausius-Clapeyron' equation

$$\frac{dP}{dT} = \frac{\Delta S_{\text{trans}}}{\Delta V_{\text{trans}}}$$

where dP/dT represents the slope of the pressure vs temperature curve when two phases coexist in equilibrium; ΔS_{trans} and ΔV_{trans} are the entropy and volume changes for the phase transition, respectively.

For cases of change of state of aggregation of a pure substance, considering that ΔS_{trans} and ΔV_{trans} are approximately constant in a small temperature range and if the vapor in equilibrium behaves like an ideal gas, the previous equation is transformed to:

Change of solid-liquid state: $\frac{P - P^\circ}{T_{\text{fus}} - T^\circ_{\text{fus}}} = \frac{\Delta H^\circ_{\text{fus}}}{\Delta V^\circ_{\text{fus}}}$ Change of liquid-vapor state: $\ln \left(\frac{P}{P^\circ} \right) = -\frac{\Delta H^\circ_{\text{vap}}}{R} \cdot \left(\frac{1}{T_{\text{eb}}} - \frac{1}{T^\circ_{\text{eb}}} \right)$

where $T^\circ_{\text{fus}} = 273,15 \text{ K}$ and $T^\circ_{\text{eb}} = 373,15 \text{ K}$ are the melting and boiling temperatures at standard pressure $P^\circ = 1,00 \text{ bar} = 100 \text{ kPa}$, respectively; $\Delta H^\circ_{\text{fus}}$ and $\Delta H^\circ_{\text{vap}}$ are the standard enthalpies of fusion and vaporization; $\Delta V^\circ_{\text{fus}}$ is the volume change that occurs when fusion occurs.

	$\Delta H^\circ_f / \text{kJ} \cdot \text{mol}^{-1}$	$S^\circ / \text{J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	$\rho / \text{g} \cdot \text{dm}^{-3}$
H ₂ O(s)	−291,8 (25 °C)	43,2 (0 °C)	917
H ₂ O(l)	−285,8 (25 °C)	65,2 (0 °C) 69,9 (25 °C) 86,8 (100 °C)	1000
H ₂ O(g)	−241,8 (25 °C)	196,9 (100 °C)	0,598

CÓDIGO:

13.2. Calculate $\Delta H^\circ_{\text{fus}}$, $\Delta S^\circ_{\text{fus}}$ (in J/(mol·K)), $\Delta V^\circ_{\text{fus}}$ (in dm³/mol), $\Delta H^\circ_{\text{vap}}$ and $\Delta S^\circ_{\text{vap}}$ (in J/(mol·K)) for water from the thermodynamic data above.

13.3. Using the above results and the Clausius-Clapeyron equations for state changes, calculate the melting temperatures T_{fus} and boiling temperatures T_{eb} (in °C) of water at 74.7 kPa.

The previous results, therefore, do not justify the existence of ice under these conditions. The students became more thoughtful when they noticed that the weather station on top of the mountain indicated 100 kPa, 5.0 °C and a relative humidity of 80%.

13.4. To be sure of all his reasoning, Caesar calculated the spontaneity of the ice-melting process under these conditions: what result did he obtain?

You must calculate $\Delta G^\circ_{\text{fus}}$ (5 °C) in kJ/mol.

Then it occurred to Erik that perhaps there was no water in the mountains because it was evaporating. Just at that moment, a strong wind began to blow, the clouds loaded with moisture collided with the mountains, and it started to rain, even from the bottom up, a product of the orography. He realized that for condensation to occur at a given temperature, the air must first be saturated with water vapor; the relative humidity must be $H_r = 100\%$ inside the cloud.

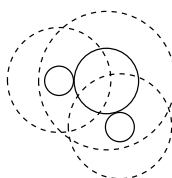
13.5. Calculate the pressure of water vapor at 5.0 °C when the air is saturated with moisture $P_{\text{H}_2\text{O}}^{\text{sat}}$ in kPa.

13.6. Help Erik calculate how much water gas (in g) is contained in 10 km³ of air around Mount Sántis when the relative humidity is 80% at 5.0°C. Consider $P_{\text{H}_2\text{O}}^{\text{sat}} = 0.50$ kPa if he failed to solve **13.5**.

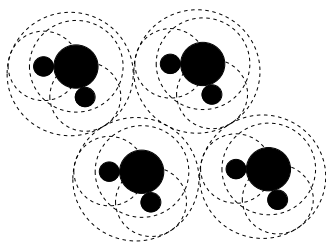
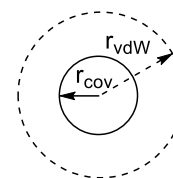
13.7. Condensation also requires a nucleus or surface where the molecules that “escape” from the gas phase are deposited. How much water per unit area is contained in a monolayer (layer one molecule thick) covering the condensation nucleus? Assume that the water molecules are spheres of radius $r = 179$ pm and are arranged as compactly as possible.

Effective radius of a water molecule in constant motion

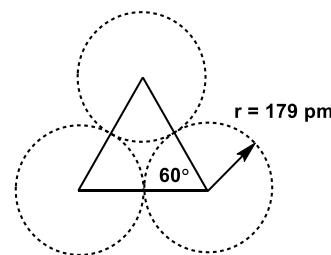
$$r_{\text{H}_2\text{O}}^{\text{eff}} \approx \frac{r_{\text{H}}^{\text{cov}} + r_{\text{H}}^{\text{vdW}} + r_{\text{O}}^{\text{cov}} + r_{\text{O}}^{\text{vdW}}}{2} = 179 \text{ pm}$$



Covalent and
van der Waals
radii



Arrangement of water molecules
in a compact monolayer



13.8. During his search, Erik found a curious fact in the chapter on chemical equilibrium: “In acidic aqueous solution, hydronium ions (H_3O^+), the Zundel cation (H_5O_2^+ , $\text{H}_3\text{O}^+ \cdot \text{H}_2\text{O}$, $\text{H}^+ \cdot 2\text{H}_2\text{O}$), the Eigen cation (H_9O_4^+ , $\text{H}_3\text{O}^+ \cdot 3\text{H}_2\text{O}$) and the Reed-Stoyanov cation ($\text{H}_{13}\text{O}_6^+$, $\text{H}^+ \cdot 6\text{H}_2\text{O}$) are present”. Although this did not help solve his questions, he represented the structures of these four species for fun, marking the relevant hydrogen bonds with dashed lines. Can you do it, too? Hint: All structures have some element of symmetry.

CÓDIGO:

13.9. César likes to experiment, and at lunchtime, he poured some of the contents of the salt shaker out the window. When he came out, he found that part of the ice outside had melted, exposing a portion of the ground. What temperature is required for water to freeze again? A saturated sodium chloride solution in water at 5.0 °C contains 26.3% by mass. $M(\text{NaCl}) = 58.44 \text{ g/mol}$

$$K_{\text{cryo}} = \frac{R \cdot M_{\text{dis}} \cdot T_{\text{fus}}^2}{1000 \cdot \Delta H_{\text{fus}}^{\circ}}$$

13.10. While reflecting, César remembered his visit to the Faculty of Chemistry and the debates with the professors of Thermodynamics and Kinetics, José Manuel Nieto and Rubén Álvarez and at that time, the mystery was solved! Can you infer why there is ice on Mount Sántis in summer?

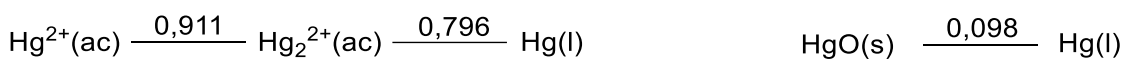
Question 14. Tasty electrochemistry! (15% of the total; 11th and 12th grades)

14.1	14.2	14.3	14.4	14.5	14.6	14.7	14.8	14.9	14.10	14.11	14.12	Total	Total
4	3	2	5	5	3	7	9	8	5	5	4	60	15%

One morning, the Ibero-American contestants Mario and Héctor studied various electrochemical systems. They had a silver chloride reference electrode (AgCl/Ag , $E^{\circ} = 0.222 \text{ V}$) and a digital multimeter. They decided to measure the potential of another electrode that had been prepared by the laboratory professor using mercury, mercurous sulfate, and a saturated solution of potassium sulfate since they could not find the value in the library books.

14.1. The electrochemical cell formed by the union of both electrodes had an EMF of 393 mV, with the silver chloride electrode acting as the anode. Represent the thermodynamic notation of the cell and calculate the unknown electrode potential $E[\text{Hg}_2\text{SO}_4(\text{s})/\text{Hg}(\text{l}), \text{SO}_4^{2-}(\text{ac})]$.

14.2. The presence of the mercurous ion caught Mario's attention, who believed that this ion was not stable in an aqueous solution. He demonstrates that the previous reasoning is incorrect, using the Latimer diagram of mercury.



14.3. Mario was convinced that the mercury(I) concentration had to be very small and that mercurous sulfate should be poorly soluble in water. How could you prove this hypothesis simply in the laboratory?

14.4. Héctor then found a note from the professor that the electrode had been prepared using a saturated solution of potassium sulfate, which has 10.0% salt by mass and a density of 1.08 g/cm³. How can the solution be prepared and what is its molar concentration? $M(\text{K}_2\text{SO}_4) = 174,27 \text{ g/mol}$.

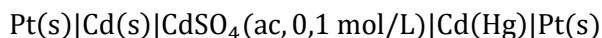
14.5. Then, Héctor decided to check the previous result. He poured a 5.00 mL aliquot of a freshly prepared saturated potassium sulfate solution into a beaker and added an excess concentrated barium chloride solution. After letting it rest, he filtered out all the white solid formed, dried it and determined that its mass was 720 mg. What is the new concentration value?

14.6. Mario observed that a tray with elemental sulfur had been placed on the work table. This is a standard procedure to collect accidentally spilled mercury residue. Because?

14.7. After examining all the data, Mario calculated the concentration of mercury(I) ions in the solution and the solubility product constant of mercurous sulfate. What values did he obtain? If I cannot solve **14.1**, consider that the potential of the mercury electrode was 0.620 V.

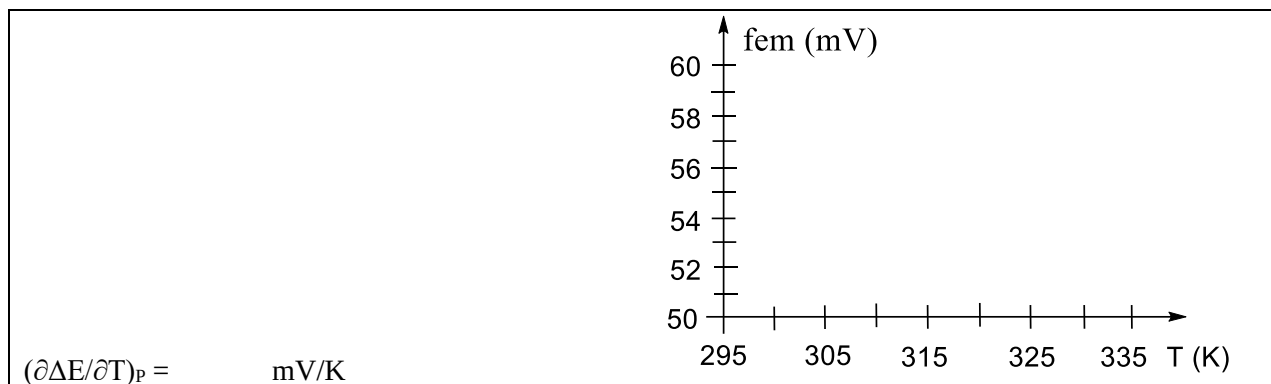
CÓDIGO:

There was also cadmium in the laboratory, and the students decided to study the formation reaction of an amalgam of this metal, for which they measured the EMF of a new cell.



T(°C)	23,5	30,3	34,6	39,2	44,0	48,6	54,9	60,4
FEM (mV)	52	53	54	55	56	57	59	60

14.8. Using the data provided, complete the ΔE vs T graph and calculate the cell temperature coefficient $(\partial\Delta E/\partial T)_P$ (the slope of the curve), assuming that the relationship between the variables is linear.



14.9. Calculate the value of the indicated thermodynamic variables at 298.15 K, with two significant figures.

$$\Delta S = -z \cdot F \cdot (\partial\Delta E/\partial T)_P$$

You must calculate ΔE (in mV), ΔG (in kJ/mol), ΔS (in J/(mol·K)) and ΔH (in kJ/mol). $T = 298,15 \text{ K}$

After work, the students drank carbonated soda. While they were resting, Héctor wondered why, if soda cans are made of aluminum, an active metal, no chemical reaction occurs. This question made them search the Internet, where they found the information they needed.

14.10. The typical acidulant in carbonated soft drinks is phosphoric acid. What is the acid concentration capable of giving the drink an acidity of $\text{pH} = 3.0$? $\text{pK}_{a1} = 2,15$; $\text{pK}_{a2} = 7,20$; $\text{pK}_{a3} = 12,35$

14.11. Show that the reaction between metallic aluminum and the acid in the soft drink is spontaneous at room temperature. $E^\circ(\text{Al}^{3+}/\text{Al}) = -1.66 \text{ V}$. You must calculate ΔE (in V) and ΔG (in kJ/mol)

14.12. In reality, the aluminum in soda cans is covered with a layer of complex composition that protects it against oxidation due to acid solution. What maximum value can the dissolved aluminum concentration take within the layer? $\text{pK}_{ps}[\text{Al}(\text{OH})_3] = 32.97$.

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gases' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamic	$Q = \Delta U + W$
Zeroth in Celsius's scale	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs's energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Vapor pressure depression	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Volume of the sphere	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Area of the sphere	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Area of the circle	$A_{\text{sp}} = \pi \cdot r^2$	Light speed	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Length of the circumference	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Force of gravity	$F_g = m \cdot g$	Gravity acceleration on the Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$ $1 \text{ bar} = 100 \text{ kPa}$ $1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$ $1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$ $1 \text{ dm}^3 = 1 \text{ L}$ $1 \text{ mL} = 1 \text{ cm}^3$ $1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3		4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																		2 He 4.003
2	3 Li 6.941	4 Be 9.012												5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305												13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80	
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29	
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	*	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207,2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	**	104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]